Sharp translation monotonicity for the polygamma – logarithmic-mean difference

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Abstract

We prove the two necessary and sufficient parameter ranges in Conjecture 5.1 of Qi's 2009 survey (also Conjecture 5.1 of its 2014 published version). A Laplace-transform difference has exactly one change of sign. Comparing its vanishing power moment against the polygamma kernel proves both sufficient ranges, including their endpoints. Small-argument and large-argument asymptotics prove necessity. The sufficient inequalities also hold for arbitrary nonconstant bounded positive random variables translated by a common parameter.

Task 1: the source problem

For $a, b > 0$ with $a \neq b$, use Qi's parameter convention

$$L_p(a,b) = \begin{cases} \left(\dfrac{b^{p+1}-a^{p+1}}{(p+1)(b-a)}\right)^{1/p}, & p \neq -1, 0, \\ \dfrac{b-a}{\log b - \log a}, & p = -1, \\ \exp\left(\dfrac{b\log b - a\log a}{b-a} - 1\right), & p = 0. \end{cases}$$

This is the generalized logarithmic mean, despite the word "Lehmer" in the MathDB title. In particular, it is the power mean of a uniform random variable on $[a, b]$. Let $i \geq 0$ be an integer and put $h_i(y) = (-1)^i \psi^{(i)}(y)$, where $\psi^{(0)} = \psi$.

Theorem 1. *Fix distinct real s, t. On $x > -\min\{s, t\}$, define*

$$f_{i,p,s,t}(x) = h_i\big(L_p(x+s, x+t)\big) - \frac{1}{t-s}\int_s^t h_i(x+u)\,\mathrm{d}u.$$

Then $f_{i,p,s,t}$ is increasing if and only if $p \leq -i-1$, and decreasing if and only if $p \geq -i$. In the respective ranges its derivative is strictly positive and strictly negative everywhere. Thus the same characterization applies whether "increasing" and "decreasing" are taken strictly or weakly.

The source locations are [1, Conjecture 5.1, p. 16] and [2, Conjecture 5.1, p. 162]. The conjecture concerns translation in x, with s, t held fixed. We prove necessity for every such fixed pair, not just uniformly over all pairs.

A comparison lemma

Lemma 1. *Let X be a nonconstant random variable with $0 < a \le X \le b < \infty$. For real $p < 1$, define*

$$m = \begin{cases} (\mathbb{E}X^p)^{1/p}, & p \neq 0, \\ \exp(\mathbb{E}\log X), & p = 0, \end{cases} \qquad K = m^{1-p}\mathbb{E}X^{p-1}.$$

Then $K > 1$, and

$$D(u) = \mathbb{E}e^{-uX} - Ke^{-um}$$

has exactly one zero u_0 on $(0,\infty)$, is negative before u_0, and positive after u_0. Moreover,

$$\int_0^\infty u^{-p}D(u)\,\mathrm{d}u = 0. \tag{1}$$

If w is strictly increasing and the relevant integrals converge absolutely, then

$$\int_0^\infty u^{-p}D(u)w(u)\,\mathrm{d}u > 0.$$

The inequality reverses if w is strictly decreasing.

Proof. For $p < 0$ or $0 < p < 1$, apply strict Jensen to $Y = X^p$ and the strictly convex function $y^{(p-1)/p}$:

$$\mathbb{E}X^{p-1} > (\mathbb{E}X^p)^{(p-1)/p} = m^{p-1}.$$

For $p = 0$, strict Jensen gives $\mathbb{E}X^{-1} > \exp(-\mathbb{E}\log X) = m^{-1}$. Thus $K > 1$.

The function $G(u) = e^{um}D(u) = \mathbb{E}e^{(m-X)u} - K$ is strictly convex, since

$$G''(u) = \mathbb{E}\big((m-X)^2e^{(m-X)u}\big) > 0.$$

Here $G(0) = 1 - K < 0$. Since the power mean lies strictly between the essential infimum and essential supremum of X, there are $\epsilon, \delta > 0$ with $\Pr(X \le m - \epsilon) \ge \delta$. Consequently $G(u) \ge \delta e^{\epsilon u} - K \to \infty$. Strict convexity now gives precisely one positive zero, with the asserted signs.

Since $1 - p > 0$, integrating the two nonnegative terms separately gives

$$\int_0^\infty u^{-p}D(u)\,\mathrm{d}u = \Gamma(1-p)\big(\mathbb{E}X^{p-1} - Km^{p-1}\big) = 0.$$

Subtract $w(u_0)$ times this identity. For increasing w, the product $D(u)(w(u) - w(u_0))$ is strictly positive except at u_0. For decreasing w it is strictly negative. This proves the comparison. □

Proof of the sufficient ranges

By symmetry assume $s < t$ and let U be uniform on $[s,t]$. For fixed $x > -s$ set $X = x + U$ and $m(x) = L_p(x+s, x+t)$. Differentiating the power-mean identity, also at $p = 0$, gives

$$m'(x) = m(x)^{1-p}\mathbb{E}X^{p-1} = K. \tag{2}$$

The standard positive series and Laplace integral are

$$h_i'(y) = (i+1)!\sum_{n=0}^{\infty}(y+n)^{-i-2} = \int_0^\infty \frac{u^{i+1}}{1-e^{-u}}e^{-yu}\,\mathrm{d}u. \tag{3}$$

They hold for every $i \geq 0$ and $y > 0$; see, for example, [3, equation (1.2)]. Hence

$$f'(x) = Kh_i'(m) - \mathbb{E}h_i'(X) = -\int_0^\infty u^{-p} D(u) w_{i,p}(u)\,\mathrm{d}u, \qquad w_{i,p}(u) = \frac{u^{i+1+p}}{1-e^{-u}}. \tag{4}$$

All integrals converge absolutely: near zero the last integrand is $O(u^i)$, and at infinity $a = x + s > 0$ supplies exponential decay.

Put $\alpha = i + 1 + p$. A direct logarithmic differentiation yields

$$\frac{w_{i,p}'(u)}{w_{i,p}(u)} = \frac{\alpha}{u} - \frac{1}{e^u - 1}.$$

If $p \leq -i - 1$, then $\alpha \leq 0$ and this is strictly negative. Lemma 1 and (4) imply $f'(x) > 0$. If $-i \leq p < 1$, then $\alpha \geq 1$, and $e^u - 1 > u$ shows that the logarithmic derivative is strictly positive. The same lemma gives $f'(x) < 0$. This includes $p = 0$ when appropriate.

For completeness, suppose $p \geq 1$. Then $m \geq \mathbb{E}X$ and $K \leq 1$. For $p > 1$, the second inequality follows from Jensen applied to $y^{(p-1)/p}$; for $p = 1$, $K = 1$. By (3), h_i' is positive, strictly decreasing, and strictly convex. Consequently

$$Kh_i'(m) \leq h_i'(m) \leq h_i'(\mathbb{E}X) < \mathbb{E}h_i'(X),$$

which again gives $f'(x) < 0$.

Necessity for every fixed interval

Translate x so that $a = x + s \downarrow 0$ is the left endpoint, and put $d = t - s > 0$, $z = a + d/2$. All parameters d, p, i are fixed below. Expansion of the defining integral of the mean about its midpoint gives

$$L_p(a, a+d) = z + \frac{(p-1)d^2}{24z} + O(z^{-3}), \qquad z \to \infty. \tag{5}$$

Indeed $\mathbb{E}(z+V)^p = z^p(1 + p(p-1)d^2/(24z^2) + O(z^{-4}))$ for V uniform on $[-d/2, d/2]$; at $p = 0$ expand $\mathbb{E}\log(z+V)$ instead. The integral test in (3), also after differentiation, gives

$$h_i'(z) = i!z^{-i-1} + O(z^{-i-2}), \qquad h_i''(z) = -(i+1)!z^{-i-2} + O(z^{-i-3}).$$

Taylor expansion, with the higher derivative bounds from the same series, therefore proves

$$f_{i,p,s,t}(x) = \frac{i!(i+p)d^2}{24z^{i+2}} + O(z^{-i-3}). \tag{6}$$

In particular $f(x) \to 0$ for every real p. If $p < -i$, this expansion is negative for sufficiently large x. A decreasing function with limit zero must be nonnegative everywhere. Thus decreasing monotonicity requires $p \geq -i$.

An increasing function with limit zero must be nonpositive everywhere. We show that $p > -i - 1$ instead makes f positive near $a = 0$. First suppose $i \geq 1$. The recurrence for the polygamma function gives $h_i(y) = -i!y^{-i-1} + O(1)$ as $y \downarrow 0$, with a bounded remainder on every fixed bounded positive interval. On integration,

$$\mathbb{E}h_i(X) = -\frac{(i-1)!}{d}a^{-i} + O(1). \tag{7}$$

For $-i - 1 < p < -1$ the mean satisfies

$$m \sim \big((-p-1)d\big)^{-1/p} a^{1+1/p}.$$

The inequality $p > -i-1$ implies $(i+1)(1+1/p) < i$, so $h_i(m) = o(a^{-i})$. For $p = -1$, the exact logarithmic-mean formula gives $h_i(m) = O((\log(1/a))^{i+1}) = o(a^{-i})$. For $p > -1$, the mean tends to a strictly positive constant, so $h_i(m)$ stays bounded. Equation (7) proves in all cases

$$f(x) \sim \frac{(i-1)!}{d} a^{-i} > 0.$$

Finally, for $i = 0$ the excluded range is $p > -1$. Here

$$m \longrightarrow \begin{cases} d(p+1)^{-1/p}, & p \neq 0, \\ d/e, & p = 0, \end{cases} \qquad \mathbb{E}\psi(X) = \frac{\log\Gamma(a+d) - \log\Gamma(a)}{d} = \frac{\log a}{d} + O(1).$$

Thus $f(x) = -(\log a)/d + O(1) \to +\infty$, which also excludes increasing monotonicity. This completes the proof of Theorem 1.

Corollary 1. *Let U be any nonconstant bounded real random variable. Define $m_p(x)$ as the power mean of $x + U$, including its geometric value at $p = 0$. On $x > -\operatorname{ess\,inf} U$, the function*

$$h_i(m_p(x)) - \mathbb{E}h_i(x+U)$$

has strictly positive derivative if $p \leq -i-1$, and strictly negative derivative if $p \geq -i$.

Proof. The sufficient-range proof uses only boundedness, positivity, and nonconstancy. Its differentiation and integral interchange are valid locally on the stated domain by uniform positive lower bounds on $x + U$. The uniform distribution was needed only for the endpoint necessity argument above. □

Follow-up tasks

These tasks follow the full solution above. They are proposed directions, not additional completed problems or independently screened open claims.

Task 2: Determine the parameters for which f or $-f$ is completely monotonic, as suggested in Qi's Remark 5.8. Start with the two boundary parameters $p = -i$ and $p = -i-1$.

Task 3: For $-i-1 < p < -i$, determine the number of zeros and stationary points of f. In particular, prove or disprove uniqueness of its zero and of its global minimum, and obtain bounds for their locations.

Task 4: Determine the sharp monotonicity ranges when the uniform law is replaced by a beta density on $[s, t]$. Identify how endpoint behavior affects necessity beyond the sufficient ranges in Corollary 1.

Task 5: Replace $\psi^{(i)}$ by the q-polygamma function for $0 < q < 1$, and determine the translation-monotonicity ranges for the same means L_p. Account for the discrete Laplace measure and the limit $q \uparrow 1$.

Status and verification

The live MathDB record and a dated literature audit accompany this file. No matching prior claim was located in the sources recorded in that audit; this is a bounded literature check, not a guarantee about all public work. The proof above is analytic. The accompanying script checks derivative signs and compares separately computed expressions at prescribed inputs; those computations are supplementary and do not replace the proof.

References

[1] F. Qi, *Bounds for the ratio of two gamma functions—From Gautschi's and Kershaw's inequalities to completely monotonic functions*, arXiv:0904.1049v1 (2009), Conjecture 5.1. https://arxiv.org/abs/0904.1049.

[2] F. Qi, *Bounds for the Ratio of Two Gamma Functions: from Gautschi's and Kershaw's Inequalities to Complete Monotonicity*, Turkish Journal of Analysis and Number Theory **2**(5) (2014), 152–164. https://doi.org/10.12691/tjant-2-5-1.

[3] Z.-H. Yang and S.-Z. Zheng, *Monotonicity of a mean related to polygamma functions with an application*, Journal of Inequalities and Applications (2016), 216. https://doi.org/10.1186/s13660-016-1155-4.